

6	(a) reference to input (voltage) and output (voltage) there is no time delay between change in input and change in output	B1 B1	[2]
	<i>or</i>		
	reference to rate at which output voltage changes infinite rate of change (of output voltage)	(B1) (B1)	
(b)	(i) $2.00/3.00 = 1.50/R$	C1	
	<i>or</i>		
	$V_+ = (3.00 \times 4.5)/(2.00 + 3.00) = 2.7$ $2.7 = 4.5 \times R/(R + 1.50)$	(C1)	
	resistance = $2.25 \text{ k}\Omega$	A1	[2]
	(ii) 1. correct symbol for LED two LEDs connected with opposite polarities between V_{OUT} and earth	M1 A1	[2]
	2. below 24°C , $R_T > 1.5 \text{ k}\Omega$ or resistance of thermistor increases/high	B1	
	$V_- < V_+$ or V_- decreases/low (must not contradict initial statement)	M1	
	V_{OUT} is positive/+5 (V) <u>and</u> LED labelled as 'pointing' from V_{OUT} to earth	A1	[3]